Coherence-Controlled Gram Quadratic Bounds and Sharpness

Liang Wang
School of Mathematics and Statistics
Huazhong University of Science and Technology
Wuhan, China
liang.wang@hust.edu.cn

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Abstract

We prove a sharp structural envelope for a weighted Gram quadratic in a complex Hilbert space. Given $g = \sum_i \lambda_i v_i$, put $a_i = |\lambda_i|\|v_i\|$, $D = \sum_i a_i^2$, and $L = \sum_i a_i$. If μ is the maximum normalized absolute inner product over distinct active vectors, with $\mu = 0$ when there is at most one, then

$$\big|\|g\|^2 - D\big| \le \mu(L^2 - D).$$

This gives a nonnegative two-sided bound. When $D > 0$, the normalized form uses $\kappa = L^2/D$, where $1 \le \kappa \le |\mathcal{A}|$, and $\mu(\kappa - 1) < 1$ certifies noncancellation. The exact independent and global support radii of TPC-249 inherit the bounds after square roots and budget factors. Positive equicorrelation and negative two-vector correlation show that the upper and signed-lower coefficients are universally sharp. Simplex and rational collinear cancellation show that the nonnegative floor is necessary. Identical marginal norms and weights can yield squared norm four or zero, so marginals alone cannot improve the L^2 upper endpoint or force a positive lower bound. These conclusions are finite structural geometry; no actual V59 coherence asymptotic or arithmetic saving is proved.

1 Introduction and source lock

Weighted sums in a shared Hilbert-space lane retain cancellation that is invisible after the summands are replaced by separate marginal copies. The exact object is the Gram quadratic, but its off-diagonal entries may be hard to evaluate on an arithmetic source. This note isolates what follows from one coarse statistic—the active coherence—without changing the shared-lane model.

TPC-249 supplies the literal contraction and its exact quadratic identity [1]:

$$g_c = \sum_b \lambda_{cb} v_{cb}, \qquad \|g_c\|^2 = \lambda_c^* G_c \lambda_c, \qquad (G_c)_{bb'} = \langle v_{cb}, v_{cb'} \rangle. \tag{1}$$

The inner product is conjugate-linear in the first slot and linear in the second. Thus the weights in g_c are not conjugated. We import no bound for the entries of the literal V59 matrices G_c.

Our contribution has four parts. First, we bound the difference between the Gram quadratic and its diagonal using coherence and weighted ℓ^1 mass. Second, we normalize the result by the diagonal energy and identify a strict noncancellation criterion. Third, we transfer the estimate to both exact TPC-249 budget geometries. Fourth, we audit every universal constant and the zero floor with positive-semidefinite Gram matrices. The last step is essential: marginal data permit both full alignment and exact cancellation.

2 Definitions and edge cases

Let $\mathcal{H}$ be a complex Hilbert space, let I be finite, and choose $\lambda_i \in \mathbb{C}$ and $v_i \in \mathcal{H}$. Define

$$g := \sum_{i \in I} \lambda_i v_i, \qquad a_i := |\lambda_i| \, \|v_i\|, \qquad \mathcal{A} := \{i \in I : a_i > 0\}, \tag{2}$$

and

$$D := \sum_{i \in I} a_i^2, \qquad L := \sum_{i \in I} a_i. \tag{3}$$

The coherence definition must cover the empty-pair cases. We set

$$\mu := \begin{cases} 0, & |\mathcal{A}| \le 1, \\ \max\limits_{\substack{i,j \in \mathcal{A} \\ i \ne j}} \dfrac{|\langle v_i, v_j \rangle|}{\|v_i\| \, \|v_j\|}, & |\mathcal{A}| \ge 2. \end{cases} \tag{4}$$

Every denominator in the second line is positive. Cauchy's inequality gives $0 \le \mu \le 1$.

If $D = 0$, every a_i vanishes. Hence every vector $\lambda_i v_i$ vanishes, $g = 0$, $L = 0$, and the unnormalized bounds below reduce to equality. We do not define κ in this case. Only when $D > 0$ do we put

$$\kappa := \frac{L^2}{D}. \tag{5}$$

This separation prevents a hidden division by zero and makes the singleton case automatic: if $|\mathcal{A}| = 1$, then $\mu = 0$ and $\kappa = 1$.

3 The coherence envelope

Theorem 3.1 (coherence-controlled Gram quadratic)**.** *With the definitions above,*

$$\Big| \|g\|^2 - D \Big| \le \mu (L^2 - D). \tag{6}$$

Consequently,

$$\max\{D - \mu(L^2 - D), 0\} \le \|g\|^2 \le D + \mu(L^2 - D). \tag{7}$$

If $D > 0$, then

$$D\,[1 - \mu(\kappa - 1)]_+ \le \|g\|^2 \le D\,[1 + \mu(\kappa - 1)], \qquad 1 \le \kappa \le |\mathcal{A}|. \tag{8}$$

In particular, $\mu(\kappa - 1) < 1$ implies $g \ne 0$.

Proof. Conjugate linearity in the first slot gives

$$\|g\|^2 = \sum_{i,j \in I} \overline{\lambda_i} \lambda_j \langle v_i, v_j \rangle, \tag{9}$$

$$\|g\|^2 - D = \sum_{\substack{i,j \in I \\ i \ne j}} \overline{\lambda_i} \lambda_j \langle v_i, v_j \rangle. \tag{10}$$

Terms with an inactive index vanish. For distinct active indices,

$$\Big| \overline{\lambda_i} \lambda_j \langle v_i, v_j \rangle \Big| \le \mu a_i a_j.$$

The triangle inequality therefore yields

$$\Big|\|g\|^2 - D\Big| \le \mu \sum_{i\neq j} a_i a_j = \mu \left[\left(\sum_i a_i \right)^2 - \sum_i a_i^2 \right],$$

which is (6). The signed lower estimate from (6), intersected with $\|g\|^2 \ge 0$, gives (7).

Suppose $D > 0$. Dividing (7) by D gives the first part of (8). Since the active a_i are positive,

$$D \le L^2 \le |\mathcal{A}| D.$$

The first inequality follows by retaining the diagonal terms in L^2; the second is Cauchy–Schwarz on the active coordinates. Thus $1 \le \kappa \le |\mathcal{A}|$. Finally, $\mu(\kappa - 1) < 1$ makes the normalized lower endpoint strictly positive, so $\|g\|^2 > 0$. □

Remark 3.2 (What the parameters record). The factor $\kappa - 1 = (L^2 - D)/D$ measures how much weighted off-diagonal mass is available relative to diagonal energy. Coherence alone does not determine the phases of the cross terms; it controls their combined magnitude. The product $\mu(\kappa - 1)$ is therefore the natural dimensionless cancellation budget in this argument.

4 Inheritance by the TPC-249 support radii

For each shared lane c, apply the preceding definitions to the family $(\lambda_{cb}, v_{cb})_b$ and write

$$B_c^- := \big[D_c - \mu_c(L_c^2 - D_c)\big]_+, \qquad B_c^+ := D_c + \mu_c(L_c^2 - D_c). \tag{11}$$

These quantities remain defined when $D_c = 0$; no κ_c is then formed.

Corollary 4.1 (independent and global budget envelopes)**.** *For independent lane radii $\rho_c \ge 0$, the exact TPC-249 radius obeys*

$$\sum_c \rho_c \sqrt{B_c^-} \le R_{\text{ind}} := \sum_c \rho_c \|g_c\| \le \sum_c \rho_c \sqrt{B_c^+}. \tag{12}$$

For one global direct-sum budget $\rho \ge 0$, its exact radius obeys

$$\rho \sqrt{\sum_c B_c^-} \le R_{\text{glob}} := \rho \sqrt{\sum_c \|g_c\|^2} \le \rho \sqrt{\sum_c B_c^+}. \tag{13}$$

Proof. Theorem 3.1 gives $B_c^- \le \|g_c\|^2 \le B_c^+$ lane by lane. For (12), take square roots, multiply by ρ_c, and sum. For (13), first sum the squared-norm bounds and then apply the increasing square-root function and the factor ρ. □

The two formulas reflect different uncertainty domains and are not interchangeable. More importantly, they are conditional structural envelopes: using them arithmetically requires source-specific control of D_c, L_c, and μ_c. This paper provides no such V59 asymptotic.

5 Sharpness and the marginal obstruction

The next propositions concern universal constants and the nonnegative floor. They do not claim that every arbitrary tuple $(\mu, \kappa, |\mathcal{A}|)$ admits a saturating family.

Proposition 5.1 (upper coefficient one is sharp)**.** *No universal constant smaller than one can replace the coefficient of $\mu(L^2 - D)$ in the upper bound.*

Proof. Fix $m \geq 2$ and $0 \leq \mu \leq 1$. Let G have diagonal entries one and every off-diagonal entry μ. Its eigenvalues are $1 - \mu$ with multiplicity $m - 1$ and $1 + (m - 1)\mu$ once. Hence G is positive semidefinite and is the Gram matrix of unit vectors. With every weight equal to one,

$$\lambda^* G \lambda = m + \mu(m^2 - m) = D + \mu(L^2 - D).$$

Thus the displayed coefficient is attained. □

Proposition 5.2 (signed-lower coefficient one is sharp)**.** *No universal constant smaller than one can replace the coefficient of $\mu(L^2 - D)$ in the signed lower estimate.*

Proof. For $0 \leq \mu \leq 1$, consider

$$G = \begin{pmatrix} 1 & -\mu \\ -\mu & 1 \end{pmatrix}.$$

Its eigenvalues are $1 - \mu$ and $1 + \mu$, so it is positive semidefinite. With weights $(1, 1)$,

$$\lambda^* G \lambda = 2 - 2\mu = D - \mu(L^2 - D).$$

The signed lower coefficient is attained. □

Proposition 5.3 (the nonnegative floor is necessary)**.** *The signed lower endpoint can be zero or negative while the Gram quadratic is exactly zero.*

Proof. For the zero endpoint, take the Gram matrix of a regular simplex of m unit vectors. Its off-diagonal entries are $-1/(m - 1)$, it is positive semidefinite, and the vectors sum to zero. With unit weights, $\mu = 1/(m - 1)$, $D = m$, $L = m$, and the signed lower endpoint is zero.

For a negative raw endpoint, let u be a unit vector, take $(v_1, v_2, v_3) = (u, u, -u)$, and use weights $(1, 1, 2)$. The Gram matrix

$$\begin{pmatrix} 1 & 1 & -1 \\ 1 & 1 & -1 \\ -1 & -1 & 1 \end{pmatrix}$$

is rank-one and positive semidefinite. Here $\mu = 1$, $D = 6$, $L = 4$, and

$$D - \mu(L^2 - D) = -4, \qquad g = u + u - 2u = 0.$$

Thus a valid nonnegative lower bound must apply the floor at zero. □

Proposition 5.4 (marginals do not control cancellation)**.** *Fixed weights and marginal norms alone cannot improve the universal upper endpoint L^2 and cannot imply a positive universal lower bound.*

Proof. Use weights $(1, 1)$ and two unit vectors. If the vectors are (u, u), their Gram matrix is $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ and the squared norm of the weighted sum is four, equal to L^2. If they are $(u, -u)$, their Gram matrix is $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$ and the squared norm is zero. Both Gram matrices are positive semidefinite, and the two families have identical weights and marginal norms. □

6 Exact finite verification

The machine-readable certificate contains eight exact rational fixtures. It checks upper saturation at $\mu = 1/3$, signed-lower saturation at $\mu = 2/5$, simplex cancellation, negative-raw-floor collinear cancellation, the aligned and anti-aligned same-marginal pair, $D = 0$, and a singleton active set. Every fixture uses a positive-semidefinite rational Gram matrix.

An independent checker imports no producer code. It parses only canonical rational strings, recomputes all principal minors and theorem quantities, and checks the fixed source hash and claim firewall. Its mutation suite rejects a type change, a semantic value change, a digest-rebound arithmetic promotion, and a stale digest. Both ordinary and optimized Python modes execute the same release logic; no optimization-sensitive assertion is used.

A separate deterministic stress program checks 128 weighted families of rational unit vectors, as well as the zero-data and singleton edge cases. All comparisons use exact rational arithmetic. The stress collection is labeled `NUMERICAL_FINITE_ILLUSTRATION_ONLY`: finite examples illustrate the theorem and detect implementation errors, but they are not asymptotic evidence.

7 Limitations and conclusion

The coherence envelope is useful only to the extent that the literal source provides favorable values of its inputs. TPC-250 does not estimate actual V59 coherence, does not prove a V59 Gram asymptotic, and does not turn the structural inequality into an arithmetic saving. In particular, no marginal replacement can supply the missing information, by Proposition 5.4.

Within its finite scope, the result is complete. The diagonal Gram energy D, weighted mass L, and active coherence μ give a two-sided quadratic envelope; for $D > 0$, $\mu(\kappa - 1) < 1$ gives a strict noncancellation certificate. TPC-249's independent and global radii inherit the envelope, and positive-semidefinite examples establish the universal sharpness of both coefficients and the necessity of the zero floor.

Claim firewall.

$$\begin{aligned}
\texttt{TPC250_ACTUAL_V59_COHERENCE_ASYMPTOTIC} &= \texttt{OPEN},\\
\texttt{TPC250_ARITHMETIC_ADVANCE} &= \texttt{NO},\\
\texttt{TPC250_FIXED_ATOM_CREDIT} &= 0,\\
\texttt{TPC250_L2} &= \texttt{NONE},\\
\texttt{TPC250_FULL_GATE_B} &= \texttt{OPEN},\\
\texttt{TPC250_FULL_GATE_B_STRICT_1_OVER_400} &= \texttt{UNPAID_GLOBAL},\\
\texttt{TPC250_TWIN_PRIME_RESULT} &= \texttt{NONE}.
\end{aligned}$$

No Route A claim is made.

Maximum supported claim.

`PROVED_STRUCTURAL_L1_COHERENCE_CONTROLLED_GRAM_QUADRATIC_SHARPNESS`

References

[1] Liang Wang. Sharp weighted contraction on shared gram lanes, 2026. TPC-249 project-local repository paper, August 25, 2026.